

Mode 5 — Square (centered / opt)

DP32 (8×4) → DP16 (8×4) → DP8 (8×4) → DP8 (4×4) → centered add-and-square

Step 1 — DP32 (8×4)

$$\text{DP32 } (8 \times 4) = \sum_{i=0}^{31} a_i b_i$$

Step 2 — DP16 (8×4)

$$\text{DP32 } (8 \times 4) = \underbrace{\sum_{i=0}^{15} a_i b_i}_{\text{DP16 } (8 \times 4)} + \underbrace{\sum_{i=16}^{31} a_i b_i}_{\text{DP16 } (8 \times 4)}$$

Step 3 — DP8 (8×4)

$$\text{DP32 } (8 \times 4) = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=16}^{23} a_i b_i}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=24}^{31} a_i b_i}_{\text{DP8 } (8 \times 4)}$$

Step 4 — DP8 (4×4)

$$\begin{aligned} \text{DP32 } (8 \times 4) = & 2^4 \underbrace{\sum_{i=0}^7 a_{H,i} b_i}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=0}^7 a_{L,i} b_i}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_{H,i} b_i}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=8}^{15} a_{L,i} b_i}_{\text{DP8 } (4 \times 4)} \\ & + 2^4 \underbrace{\sum_{i=16}^{23} a_{H,i} b_i}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=16}^{23} a_{L,i} b_i}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=24}^{31} a_{H,i} b_i}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=24}^{31} a_{L,i} b_i}_{\text{DP8 } (4 \times 4)} \end{aligned}$$

Step 5 — Centered add-and-square (subtract 8 from unsigned nibbles; per DP8 (4×4) add $c = 0/512/2048$ for 0/1/2 unsigned nibbles)

$$\begin{aligned} \text{DP32 } (8 \times 4) = & 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i} + b_i)^2 - \sum_{i=0}^7 a_{H,i}^2 - \sum_{i=0}^7 b_i^2 \right)}_{\text{DP8 } (4 \times 4)} \\ & + \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 ((a_{L,i} - 8) + b_i)^2 - \sum_{i=0}^7 (a_{L,i} - 8)^2 - \sum_{i=0}^7 (b_i - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\ & + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=8}^{15} (a_{H,i} + b_i)^2 - \sum_{i=8}^{15} a_{H,i}^2 - \sum_{i=8}^{15} b_i^2 \right)}_{\text{DP8 } (4 \times 4)} \\ & + \underbrace{\frac{1}{2} \left(\sum_{i=8}^{15} ((a_{L,i} - 8) + b_i)^2 - \sum_{i=8}^{15} (a_{L,i} - 8)^2 - \sum_{i=8}^{15} (b_i - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \\ & + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=16}^{23} (a_{H,i} + b_i)^2 - \sum_{i=16}^{23} a_{H,i}^2 - \sum_{i=16}^{23} b_i^2 \right)}_{\text{DP8 } (4 \times 4)} \\ & + \underbrace{\frac{1}{2} \left(\sum_{i=16}^{23} ((a_{L,i} - 8) + b_i)^2 - \sum_{i=16}^{23} (a_{L,i} - 8)^2 - \sum_{i=16}^{23} (b_i - 8)^2 + 512 \right)}_{\text{DP8 } (4 \times 4)} \end{aligned}$$

$$\begin{aligned}
& + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=24}^{31} (a_{H,i} + b_i)^2 - \sum_{i=24}^{31} a_{H,i}^2 - \sum_{i=24}^{31} b_i^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + \underbrace{\frac{1}{2} \left(\sum_{i=24}^{31} ((a_{L,i}-8) + b_i)^2 - \sum_{i=24}^{31} (a_{L,i}-8)^2 - \sum_{i=24}^{31} (b_i-8)^2 + 512 \right)}_{\text{DP8 (4}\times\text{4)}}
\end{aligned}$$

Step 6 — All components

$$\begin{aligned}
\text{PE} &= 2^4 \sum_{i=0}^7 (a_{H,i} + b_i)^2 + \sum_{i=0}^7 ((a_{L,i}-8) + b_i)^2 + 2^4 \sum_{i=8}^{15} (a_{H,i} + b_i)^2 + \sum_{i=8}^{15} ((a_{L,i}-8) + b_i)^2 \\
&+ 2^4 \sum_{i=16}^{23} (a_{H,i} + b_i)^2 + \sum_{i=16}^{23} ((a_{L,i}-8) + b_i)^2 + 2^4 \sum_{i=24}^{31} (a_{H,i} + b_i)^2 + \sum_{i=24}^{31} ((a_{L,i}-8) + b_i)^2 \\
\alpha &= 2^4 \sum_{i=0}^7 a_{H,i}^2 + \sum_{i=0}^7 (a_{L,i}-8)^2 + 2^4 \sum_{i=8}^{15} a_{H,i}^2 + \sum_{i=8}^{15} (a_{L,i}-8)^2 \\
&+ 2^4 \sum_{i=16}^{23} a_{H,i}^2 + \sum_{i=16}^{23} (a_{L,i}-8)^2 + 2^4 \sum_{i=24}^{31} a_{H,i}^2 + \sum_{i=24}^{31} (a_{L,i}-8)^2 \\
\beta &= 2^4 \sum_{i=0}^7 b_i^2 + \sum_{i=0}^7 (b_i-8)^2 + 2^4 \sum_{i=8}^{15} b_i^2 + \sum_{i=8}^{15} (b_i-8)^2 \\
&+ 2^4 \sum_{i=16}^{23} b_i^2 + \sum_{i=16}^{23} (b_i-8)^2 + 2^4 \sum_{i=24}^{31} b_i^2 + \sum_{i=24}^{31} (b_i-8)^2 \\
C &= 0 + 512 + 0 + 512 + 0 + 512 + 0 + 512 = 2048
\end{aligned}$$